

enabled device when within a designated perimeter.

Countering terror attacks

Geofencing based systems are now increasingly attracting the attention of countries in fighting the menace of terrorism attacks. This technology has been found to be very effective in countering terrorism.

For example, the UK government is planning to install high-tech digital geofence fields around iconic buildings and bridges to prevent and counter sudden terror attacks using cars and vans to plough down pedestrians in busy areas. London witnessed three such attacks this year, including an attack on the Parliament in March, and at London Bridge and a mosque in Finsbury Park. The geofence systems being developed by the UK government rely on satellites to create electronic boundaries around specific sites.

Geofencing would help connect with on-board computers in vehicles to automatically slow them down to walking pace, or to block unauthorised vehicles from entering security-sensitive areas. If a driver tries to cross the electronic boundary, the system would connect with their on-board computer and limit the vehicle's speed to a very safe level.

A UK company is working on such technology by using telematics, that is black box-style devices, to shut down a car or lorry when it has been hijacked. Sweden has already adapted the technology and vehicle manufacturers such as Scania and Volvo are in the process of conducting trials.

Similar technology is in use in drones to stop them from flying into restricted air space. It has also been proposed that government regulators should motivate drone manufacturers to factor in geofencing restrictions into unmanned aerial vehicle navigation systems that would overrule the instructions given by an unseasoned



Fig. 7: Using geofencing you can market at events without physically being there (Source: <https://prmg.com>)



Fig. 8: London saw three terror attacks this year, which led to the need for geofencing (Source: <http://stjohnstreet.co.uk>)

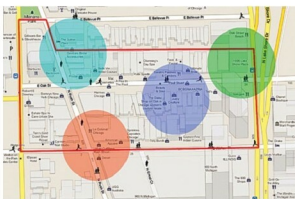


Fig. 9: Colours—an innovative feature enables users to easily differentiate between geofenced areas (Source: <https://which-50.com>)

operator, thus avoiding flight into protected airspace.

What the future holds

Today, geofencing is being used in personal productivity management, smart home control, home security, family tracking, 'if this then that' systems and quite widely in commercial businesses. And the rapid development of technology promises even more in the future.

Work is going on to develop geofencing features that integrate the virtual environment with increasing accessibility of GPS technology, giving rise to new possibilities. For example, users can assign borders and fill colours when creating or editing a geofence. They can create as many geofences as they wish and colour-code the different fences. This enables the users to differentiate between the various geofenced areas more conveniently and speedily on the map.

The platform's mapping functionality has also evolved—with the ability of a user to filter assets depending on asset status, alert status and within a last-reported range. This filtering allows the user to quickly screen out tracked assets that are not of interest at that time. This is a useful feature for users when, for example, they quickly want to see only those assets that are presently on the move or all assets that have a current alert.

We can expect developments in every sphere ranging from artificial intelligence (AI) for workstations to managing daily household tasks. As devices become more intelligent and more aspects of our home, automobiles and workplace enter the ever-growing fold of Internet-of-Things objects, we can look forward to seeing geofences applied to an increasing number of gadgets and environments.

This heightened integration is likely to result in all kinds of innovation such as desktops that switch off when their users leave for the day, tea kettles that switch on in the morning when the first tea drinkers arrive, soothing music that turns on to greet you as you arrive home, garage doors that open automatically as you reach close to home and more. Thus geofencing is likely to have a major impact on the way we work, play and live. **EFY**